

Mode 9 — Square

DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → DP8 (4×4) → add-and-square

Step 1 — DP8 (16×16)

$$\text{DP8}(16 \times 16) = \sum_{i=0}^7 a_i b_i$$

Step 2 — DP8 (16×8)

$$\text{DP8}(16 \times 16) = \underbrace{2^8 \sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8}(16 \times 8)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8}(16 \times 8)}$$

Step 3 — DP8 (16×4)

$$\text{DP8}(16 \times 16) = \underbrace{2^{12} \sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8}(16 \times 4)} + \underbrace{2^8 \sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8}(16 \times 4)} + \underbrace{2^4 \sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8}(16 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8}(16 \times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP8}(16 \times 16) = & \underbrace{2^{20} \sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8}(8 \times 4)} + \underbrace{2^{16} \sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8}(8 \times 4)} + \underbrace{2^{12} \sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8}(8 \times 4)} + \underbrace{2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8}(8 \times 4)} \\ & + \underbrace{2^8 \sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8}(8 \times 4)} + \underbrace{2^8 \sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8}(8 \times 4)} + \underbrace{2^4 \sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8}(8 \times 4)} \end{aligned}$$

Step 5 — DP8 (4×4)

$$\begin{aligned} \text{DP8}(16 \times 16) = & \underbrace{2^{24} \sum_{i=0}^7 a_{H,i}^{hi} b_i^{hh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{20} \sum_{i=0}^7 a_{L,i}^{hi} b_i^{hh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{20} \sum_{i=0}^7 a_{H,i}^{hi} b_i^{hl}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{16} \sum_{i=0}^7 a_{L,i}^{hi} b_i^{hl}}_{\text{DP8}(4 \times 4)} \\ & + \underbrace{2^{16} \sum_{i=0}^7 a_{H,i}^{lo} b_i^{hh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{12} \sum_{i=0}^7 a_{L,i}^{lo} b_i^{hh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{16} \sum_{i=0}^7 a_{H,i}^{hi} b_i^{lh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{12} \sum_{i=0}^7 a_{L,i}^{hi} b_i^{lh}}_{\text{DP8}(4 \times 4)} \\ & + \underbrace{2^{12} \sum_{i=0}^7 a_{H,i}^{lo} b_i^{hl}}_{\text{DP8}(4 \times 4)} + \underbrace{2^8 \sum_{i=0}^7 a_{L,i}^{lo} b_i^{hl}}_{\text{DP8}(4 \times 4)} + \underbrace{2^{12} \sum_{i=0}^7 a_{H,i}^{hi} b_i^{ll}}_{\text{DP8}(4 \times 4)} + \underbrace{2^8 \sum_{i=0}^7 a_{L,i}^{hi} b_i^{ll}}_{\text{DP8}(4 \times 4)} \\ & + \underbrace{2^8 \sum_{i=0}^7 a_{H,i}^{lo} b_i^{lh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^4 \sum_{i=0}^7 a_{L,i}^{lo} b_i^{lh}}_{\text{DP8}(4 \times 4)} + \underbrace{2^4 \sum_{i=0}^7 a_{H,i}^{lo} b_i^{ll}}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{ll}}_{\text{DP8}(4 \times 4)} \end{aligned}$$

Step 6 — Add-and-square (substitute each DP8 (4×4) = $\frac{1}{2} [\sum (x_i + b_i)^2 - \sum x_i^2 - \sum b_i^2]$)

$$\text{DP8}(16 \times 16) = \underbrace{2^{24} \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8}(4 \times 4)}$$

$$\begin{aligned}
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 7 — All components

$$\begin{aligned}
\text{PE} &= 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hl})^2 + 2^{16} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hl})^2 \\
&+ 2^{16} \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hh})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hh})^2 + 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{lh})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{lh})^2 \\
&+ 2^{12} \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hl})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hl})^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{ll})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{ll})^2 \\
&+ 2^8 \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{lh})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{lh})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{ll})^2 + \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{ll})^2 \\
\alpha &= 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{20} \sum_{i=0}^7 (a_{L,i}^{hi})^2 + 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{16} \sum_{i=0}^7 (a_{L,i}^{hi})^2 \\
&+ 2^{16} \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi})^2 \\
&+ 2^{12} \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi})^2 \\
&+ 2^8 \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo})^2 + \sum_{i=0}^7 (a_{L,i}^{lo})^2 \\
\beta &= 2^{24} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (b_i^{hl})^2 + 2^{16} \sum_{i=0}^7 (b_i^{hl})^2 \\
&+ 2^{16} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{12} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{16} \sum_{i=0}^7 (b_i^{lh})^2 + 2^{12} \sum_{i=0}^7 (b_i^{lh})^2 \\
&+ 2^{12} \sum_{i=0}^7 (b_i^{hl})^2 + 2^8 \sum_{i=0}^7 (b_i^{hl})^2 + 2^{12} \sum_{i=0}^7 (b_i^{ll})^2 + 2^8 \sum_{i=0}^7 (b_i^{ll})^2 \\
&+ 2^8 \sum_{i=0}^7 (b_i^{lh})^2 + 2^4 \sum_{i=0}^7 (b_i^{lh})^2 + 2^4 \sum_{i=0}^7 (b_i^{ll})^2 + \sum_{i=0}^7 (b_i^{ll})^2
\end{aligned}$$